DEPARTMENT OF COMPUTER ENGINEERING

Don Bosco Institute of Technology, Mumbai

University of Mumbai

SEMESTER V

Course Syllabus

DB25 V1 Scheme

S.No.	Course Code	Course Name	Category
1	25CE5PCC01	System Programming and Compiler Construction	PCC
2	25CE5PCC02	Artificial Intelligence	PCC
3	25CE5PCE01	Program Elective: Cloud Computing / Cyber Security / Statistics for ML & DS	PCE
4	25XX5MDMYY	MDM 02 – Android Application Development	MDM
5	25CE5OE01	Open Elective 01	OE
6	25CE5AEC02	Professional Communication Skills	AEC
7	25CE5ELC	Research Methodology	ELC
8	25CE4CEP03	Community Engagement Project – 3	CEP

Academic Year 2025–26

1. System Programming and Compiler Construction

Course Code: 25CE5PCC01 | Category: PCC

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5PCC01	**System Programming and Compiler Construction**	3	2	-	3	1	-	**4**

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	**100**
Practical/Lab		-	-	-	25	-	25	**50**
Grand Total								**150**

Pre-Requisite Course(s)	25CE4PCC01 – Theoretical Computer Science

Course Objectives & Module overview

Course Objectives: This course introduces the fundamental concepts of system software and compiler design. Students will learn the various phases of a compiler, implement language processing tools, and apply code generation and optimization techniques for efficient program translation and execution.

Module 1: Introduction to System Software and Assembler

This module introduces the role of system software in program development and execution. It covers assemblers and the design of a two-pass assembler used for translating assembly language programs.

Module 2: Macro Processor, Linkers and Loaders

This module focuses on macro processing and program loading mechanisms. Students learn how macros are expanded and how loaders and linkers prepare programs for execution.

Module 3: Lexical Analysis

This module presents the first phase of a compiler, lexical analysis. It explains token recognition using finite automata and the design of lexical analyzers.

Module 4: Syntax and Semantic Analysis

This module covers parsing techniques used to verify program structure and meaning. It introduces context-free grammars, parser design, and semantic analysis methods.

Module 5: Intermediate Code Generation

This module discusses machine-independent program representations. Students learn different intermediate code forms and their role in compiler construction.

Module 6: Code Optimization and Code Generation

This module focuses on improving code efficiency and generating target code. It introduces optimization

techniques, basic blocks, flow graphs, and code generation strategies.

Overall, the course equips computer engineering students with the knowledge and practical skills to analyze, design, and implement various phases of a compiler and system software, facilitating efficient program translation, optimization, and execution.

Course Outcomes

CO	Course Outcome
CO1	Recall the concepts of system software, assemblers, macro processors, loaders, linkers, and compiler phases.
CO2	Explain the design and working of assemblers, macro processors, lexical analyzers, parsers, and semantic analyzers.
CO3	Apply lexical analysis, parsing techniques, and LEX/YACC tools for language processing.
CO4	Analyze grammars, parsing methods, semantic rules, and intermediate code representations.
CO5	Evaluate code generation and optimization techniques for efficient program execution.
CO6	Design and implement compiler components including lexical analyzers, parsers, and code generators.

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to System Software and Assembler**		**07**
	Recall and define concepts of system software. Explain functions of assemblers, compilers, interpreters, loaders, linkers, editors, and debuggers. Describe the design and working of a two-pass assembler.		
	1.1	Concept of System Software, Goals of system software, system program and system programming	
	1.2	Introduction to various system programs: Assembler, Macro processor, Loader, Linker, Compiler, Interpreter, Device Drivers, Operating System, Editors, Debuggers	
	1.3	Assembler: Elements of Assembly Language programming, Assembly scheme, Assembler Design: Two-pass assembler design, Data structures used	
	Self-Learning Topics: *Interpreters: Working and Applications of Language Interpreters*		
2	**Module 2: Macro Processor, Linkers and Loaders**		**06**
	Recall macro definitions, macro calls, and macro expansion. Explain the design and operation of a two-pass macro processor. Describe the functions and types of loaders.		
	2.1	Macro definitions and calls, Macro expansion, Design of Two-pass macro processor, Data structures used	
	2.2	Loaders – Functions, Types (absolute, relocating, dynamic)	
	Self-Learning Topics: *Direct Linking Loader*		
3	**Module 3: Lexical Analysis**		**03**
	Recall the role of lexical analysis in compiler design. Explain the use of finite state		

	automata for token recognition. Describe the design and implementation of a lexical analyzer.		
	3.1	Role of Finite State Automata in Lexical Analysis	
	3.2	Analysis, Design of Lexical analyzer, Data structures used	
	Self-Learning Topics: *LEX Tool for Automatic Generation of Lexical Analyzers*		
4	**Module 4: Syntax and Semantic Analysis**		**14**
	Explain the role of context-free grammars in syntax analysis. Distinguish between top-down and bottom-up parsing techniques. Describe semantic analysis and syntax-directed definitions.		
	4.1	Role of Context Free Grammar in Syntax analysis	
	4.2	Types of Parsers: Top-down parser – LL(1); Bottom-up parser – SR Parser, Operator precedence parser, SLR	
	4.3	Semantic Analysis, Syntax directed definitions	
	Self-Learning Topics: *LR(0), Parser Generators and Introduction to YACC*		
5	**Module 5: Intermediate Code Generation**		**07**
	Explain the need for intermediate code. Illustrate different intermediate code representations. Describe three-address code using triples, quadruples, and indirect triples.		
	5.1	Types of Intermediate codes: Syntax tree, Postfix notation	
	5.2	Three address codes: Triples and Quadruples, Indirect triple	
	Self-Learning Topics: *Directed Acyclic Graph, Abstract Syntax Trees (AST) and Their Applications in Compilers*		
6	**Module 6: Code Optimization and Code Generation**		**08**
	Explain the need and sources of code optimization. Compare machine-dependent and machine-independent optimization techniques. Describe code generation using basic blocks and flow graphs.		
	6.1	Need and sources of optimization	
	6.2	Code optimization techniques: Machine Dependent and Machine Independent	
	6.3	Code Generation: Basic block and flow graph	
	Self-Learning Topics: *Just-In-Time (JIT) Compilation and Modern Optimization Techniques*		
		TOTAL	**45**

Laboratory Experiments (Suggested)

No.	Title and Details
1	Implementation of Two-Pass Assembler **Objective: To implement a two-pass assembler for translating assembly language programs into machine code.** **Outcome: Identify and construct symbol tables, opcode tables, and location counters.**
2	Implementation of Lexical Analyzer

	Objective: To implement the lexical analysis phase of a compiler for token identification and classification. **Outcome: Identify tokens, keywords, identifiers, and operators from source code.**
3	Implementation of Top-Down Parser **Objective: To implement a top-down parsing technique for syntax analysis of context-free grammars.** **Outcome: Apply top-down parsing methods for grammar recognition.**
4	Implementation of Bottom-Up Parser **Objective: To implement a bottom-up parser for analyzing the structure of source programs.** **Outcome: Apply shift-reduce parsing techniques for syntax analysis.**
5	Implementation of Intermediate Code Generation Phase **Objective: To generate machine-independent intermediate code representations from source programs.** **Outcome: Generate three-address code for source programs.**
6	Implementation of Code Generation Phase **Objective: To implement the code generation phase for translating intermediate code into target code.** **Outcome: Generate target code from intermediate representations.**
7	Study and Implement Experiments on LEX and YACC **Objective: To learn and apply LEX and YACC tools for automated lexical and syntax analysis.** **Outcome: Develop lexical analyzers using LEX; Construct parsers using YACC.**
8	Recognize Valid Arithmetic Expressions Using YACC **Objective: To develop a YACC-based parser for validating arithmetic expressions.** **Outcome: Design grammar rules for arithmetic expressions and implement expression validation.**

Text Books

1. A. V. Aho, R. Shethi, Monica Lam, J.D. Ullman – Compilers: Principles, Techniques and Tools, Pearson Education, Second Edition
2. J. J. Donovan – Systems Programming, Tata McGraw-Hill, 1991
3. D. M. Dhamdhere – Compiler Construction, 2nd Edition, Macmillan Publications

Reference Books

4. John R. Levine, Tony Mason & Doug Brown – Lex & YACC, O'Reilly, Second Edition
5. Leland L. Beck – System Software: An Introduction to System Programming, Pearson, Third Edition

Useful Links

6. http://www.nptelvideos.in/2012/11/compiler-design.html
7. https://www.coursera.org/lecture/nand2tetris2/unit-4-1-syntax-analysis-5pC2Z

Assessment Methodology – System Programming & Compiler Construction

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test | Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions compulsory. Q1 A or B – 10 marks Q2 A or B – 10 marks | Q3 A or B – 10 marks. Each question A and B based on same CO. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus. Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks. Based on performance and satisfactory completion of assigned laboratory work.
PR – Laboratory (25 Marks)	Practical examination based on experiments performed by the students during laboratory sessions.

2. Artificial Intelligence

Course Code: 25CE5PCC02 | Category: PCC

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5PCC02	Artificial Intelligence	3	2	-	3	1	-	4

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	100
Lab		-	-	-	25	-	25	50
Grand Total								150

Pre-Requisite Course(s)	25FE: Python Programming 25FE2PCC01: Data Structures 25CE3PCC02: Analysis of Algorithms 25CE4PCC004: Advanced Data structure and Algorithm

Course Objectives & Module overview

Course Objective:

Module 1 : **Introduction to Artificial Intelligence and Intelligent Agents**

This module introduces the foundations of Artificial Intelligence, including its evolution, major applications, and different categories of AI systems. Students learn how intelligent agents perceive and interact with their environments using rational decision-making principles. The module also covers various agent architectures and environmental characteristics that influence agent behavior.

Module 2 : **Problem-Solving Techniques in AI**

This module focuses on AI problem-solving through search and optimization techniques. Students study uninformed and informed search algorithms, heuristic-based problem solving, local optimization methods, adversarial search strategies used in game-playing, and constraint satisfaction techniques. Real-world CSP applications such as Sudoku solving, cryptarithmetic puzzles.

Module 3 : **Knowledge Representation and Reasoning**

This module introduces methods for representing knowledge in intelligent systems. Students learn propositional and first-order logic, knowledge engineering principles, inference mechanisms, and reasoning strategies. The module also covers semantic networks and ontologies for structured knowledge representation, enabling intelligent reasoning and decision-making.

Module 4 : **Fundamentals of Neural Networks**

This module provides the foundation of Artificial Neural Networks by examining biological inspiration, neuron models, network architectures, and activation functions. Students learn various learning rules, training methodologies, and different learning paradigms used in machine learning systems. The module establishes the theoretical basis for modern deep learning approaches.

Module 5 : **Associative Memory Networks**

This module focuses on memory-based neural network models that store and retrieve patterns through associative mechanisms. Students study auto-associative and hetero-associative memory systems along with classical models such as Bidirectional Associative Memory (BAM) and Hopfield Networks. The module highlights the significance of associative memory in intelligent information retrieval and pattern recognition.

Module 6 : Large Language Models and Generative AI

This module introduces the concepts and working principles of Large Language Models (LLMs) that power modern Generative AI systems. Students explore Transformer architecture, key design components, model training methodologies, and adaptation techniques used in contemporary LLMs. The module also includes case studies of leading LLMs and discusses their applications, capabilities, and evaluation approaches in real-world AI systems.

Course Outcomes

CO	Course Outcome
CO1	Recall and describe the fundamental concepts of Artificial Intelligence, intelligent agents, agent architectures, and their applications in real-world environments.
CO2	Explain search, optimization, adversarial search, and constraint satisfaction techniques to formulate and solve AI problems.
CO3	Apply knowledge representation and reasoning techniques using propositional logic, first-order logic, semantic networks, and ontologies to solve AI problems.
CO4	Analyze the architecture, learning paradigms, and training mechanisms of artificial neural networks for intelligent decision-making.
CO5	Assess the effectiveness of BAM and Hopfield networks in associative memory and pattern retrieval tasks.
CO6	Develop LLM-driven applications by applying suitable model architectures, training approaches, and prompting strategies for real-world Generative AI tasks.

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to Artificial Intelligence and Intelligent Agents**		**05**
	1.1	Introduction to AI: Definition, history, Types of AI and key applications, Turing Test	
	1.2	Agents and Environment: Definition, PEAS framework, Rational agents, Types of agents (reflex, goal-based, utility-based) and environment types	
	Self-Learning Topics: *Case studies of intelligent agents used in real-world systems (chatbots, recommendation systems, autonomous systems), Modern Intelligent Agents and Virtual Assistants, Multi-Agent Systems*		
2	**Module 2: Problem-Solving Techniques in AI**		**14**

	2.1	Uninformed Search: Depth-First Search, Breadth-First Search, Uniform Cost Search, Depth-Limited Search	
	2.2	Informed Search: Heuristic search, A* Search	
	2.3	Local Search and Optimization: Hill Climbing, Genetic Algorithms; Adversarial Search: Game Playing, Minimax Algorithm, Alpha-Beta Pruning	
	2.4	Constraint Satisfaction Problems: Cryptarithmetic problem, Sudoku Puzzle solving	
	Self-Learning Topics: Advanced Search Techniques: Bidirectional Search, AO* search.		
3	**Module 3: Knowledge Representation and Reasoning**		**09**
	3.1	Knowledge Representation: Propositional Logic, Logical Agents.	
	3.2	First-Order Logic: Syntax and Semantics, Knowledge Engineering in FOL, Inferences in FOL, Forward chaining, Backward chaining.	
	3.3	Semantic Networks and Ontology**:** Concepts and Components, Applications in AI.	
	Self-Learning Topics: *Retrieval-Augmented Generation (RAG)*		
4	**Module 4: Fundamentals of Neural Networks**		**07**
	4.1	Introduction: Biological Neural System, Artificial Neural Networks, *Neural Network Architecture,* McCulloch-Pitts Neuron, Activation Functions: Binary Step, Sigmoid, Tanh, ReLU (Introduction)	
	4.2	Learning Rules and Training : Hebbian Learning, Perceptron, Delta learning rule, Backpropagation Algorithm (Conceptual Overview)	
	4.3	Types of Learning : Supervised, Unsupervised, Semi-supervised, Reinforcement Learning	
	Self-Learning Topics: *Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Transfer Learning, Explainable AI (XAI)*		
5	**Module 5: Associative Memory Networks**		**05**
	5.1	Introduction to Associative Memory, Auto-associative and Hetero-associative Memory Networks	
	5.2	Classical Memory Models : Bidirectional Associative Memory (BAM), Hopfield Network	
	Self-Learning Topics: *Applications of Associative Memory Networks, Modern Associative Memory in Generative AI, Agentic AI (Introduction)*		
6	**Module 6: Large Language Models**		**05**
	6.1	Introduction to Large Language Models, Transformer Architecture and LLM Design : Tokenization and Embedding Layers, Positional Encoding, Self-Attention and Multi-Head Attention, Feed-Forward Networks, LLM Architectural Parameters (Layers, Hidden Dimensions, Attention Heads, Context Window)	
	6.2	LLM Training Methodologies: Self-Supervised Learning,Next Token Prediction, Pre-training, Fine-tuning, Instruction Tuning, Reinforcement	

		Learning from Human Feedback (RLHF)	
	6.3	Contemporary LLMs and Applications Case Studies of Contemporary LLMs: Comparative Analysis of Popular LLM Families, LLM Applications in Conversational AI, Knowledge Discovery, Code Generation, and Content Creation, Generative AI Applications using LLMs	
	Self-Learning Topics: Agentic AI and LLM-Based Autonomous Agents, LLM Evaluation Metrics and Benchmarking		
		TOTAL	**45**

Laboratory Experiments (Minimum 8)

No.	Title and Details
1	Case Study on AI Applications Published in IEEE/ACM/Springer or Other Prominent Journals. **Objective: To study and analyze a real-world Artificial Intelligence application published in a reputed research journal (IEEE, ACM, Springer, Elsevier, etc.).** **Outcome: Identify the AI techniques, algorithms, datasets, and evaluation methods used in the selected case study.**
2	Implementation Uninformed and Informed Search Algorithms. **Objective: Implement Uninformed Search Algorithms (Breadth-First Search, Depth-First Search, and Uniform Cost Search) and the Informed Search Algorithm (A* Search).** **Outcome: Successful implementation of BFS, DFS, UCS, and A* search algorithms.**
3	Implementation of Knowledge Representation using Propositional Logic and First-Order Logic (FOL) . **Objective:To represent knowledge using Propositional Logic and First-Order Logic.** **Outcome: To identify and represent knowledge using logical expressions and perform logical inference using Propositional Logic and FOL**
4	Design and Implementation of Semantic Networks and Ontology for Knowledge Representation **Objective:To model knowledge using Semantic Networks and Ontologies.** **Outcome:To construct Semantic Networks and simple Ontologies for representing domain knowledge and understand their role in AI-based knowledge management and reasoning.**
5	Implementation of McCulloch-Pitts Neuron and Activation Functions . **Objective:To compare the outputs of various activation functions for different inputs.** **Outcome: To identify the McCulloch-Pitts neuron model and compare the behavior of different activation functions.**
6	Implementation of a Feedforward Neural Network using Backpropagation **Objective:To understand the concepts of forward propagation, error calculation, and weight updates .** **Outcome:To train a Feedforward Neural Network using the Backpropagation algorithm and evaluate its performance for simple prediction or classification tasks**
7	Implementation of Auto-Associative and Hetero-Associative Memory Networks . **Objective:To implement associative memory models for storing and recalling patterns.** **Outcome:To identify and implement associative memory networks and demonstrate the storage and retrieval of patterns.**
8	Implementation of Bidirectional Associative Memory (BAM) **Objective:To evaluate the performance of BAM in storing and retrieving associated pattern**

	pairs. **Outcome:To store and retrieve associated pattern pairs using the BAM model.**
9	Implementation of Hopfield Network for Pattern Storage and Recall **Objective: To implement the Hopfield Network for associative memory**. **Outcome: To analyze the pattern recall capability of the Hopfield Network.**
10	Implementation of a Generative AI Chatbot using a Large Language Model . **Objective:To evaluate the performance and capabilities of the LLM for text generation and conversational AI tasks.** **Outcome:To build and evaluate a Generative AI application using a pre-trained Large Language Model (LLM).**

Assessment Methodology – Artificial Intelligence

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test | Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions are compulsory. Q1 A or B – 10 marks Q2 A or B – 10 marks | Q3 A or B – 10 marks. Each question A and B based on same CO. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions are compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus. Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks. Based on performance and satisfactory completion of assigned laboratory work.
PR – Laboratory (25 Marks)	Practical examination based on experiments performed by the students during laboratory sessions.

3. Program Elective – PCE (Choose One)

Course Code: 25CE5PCE01 | Category: PCE

Students must choose ONE of the following three elective courses:

- Cloud Computing
- Cyber Security
- Statistics for Machine Learning and Data Science

3a. Cloud Computing

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5PCE01	Cloud Computing	2	1	-	2	1	-	3

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	100
Lab		-	-	-	25	-	25	50
Grand Total								150

Pre-Requisite Course(s)	25CE4PCC03: Computer Networks

Course Objectives & Module overview

Course Objectives:

The course "Cloud Computing" is designed to provide students with a comprehensive understanding of the fundamental concepts, technologies, and applications of cloud computing in modern engineering. In today's interconnected world, cloud computing plays a pivotal role in enabling scalable, on-demand access to computational resources, storage, and applications over the internet. Emphasis will be placed on understanding how cloud computing transforms traditional telecommunication systems by providing efficient resource management, virtualization, and high availability for network services.

Module 1 This module introduces the basic concepts of cloud computing. Students learn about cloud characteristics such as on-demand access, scalability, resource pooling, and pay-as-you-go services. It covers cloud components, deployment models, benefits and challenges of cloud computing, and an overview of popular cloud platforms like AWS and Azure.

Module 2 This module Introduces virtualization concepts, virtual machines, hypervisors, and containerization technologies, including Docker and its architecture.

Module 3 This Module explains cloud service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Database as a Service (DaaS), along with their features, responsibilities, and real-world applications.

Module 4 This Module focuses on cloud security mechanisms, identity and access management(IAM), encryption, firewalls, authentication, authorization, accounting (AAA), and Single Sign-On (SSO).

Module 5 This Module Introduces distributed, parallel, grid, and quantum computing, highlighting their integration with cloud and emerging technologies like edge computing.

Module 6 This module Covers deployment, monitoring, and management of cloud resources, including virtual machines, storage, networking, load balancing, auto-scaling, and serverless computing.

Course Outcomes – Cloud Computing

CO	Course Outcome
CO1	Recall and describe the fundamental concepts, characteristics, deployment models, and service models of cloud computing. (Remembering)
CO2	Explain virtualization, containerization, cloud service models, and security mechanisms used in cloud environments. (Understanding)
CO3	Apply cloud technologies to create, configure, and manage cloud resources and applications. (Applying)
CO4	Analyze cloud deployment models, virtualization approaches, and security solutions for various application requirements. (Analyzing)
CO5	Evaluate cloud platforms, monitoring tools, and security practices based on performance and scalability. (Evaluating)
CO6	Design, deploy, and demonstrate cloud-based applications by integrating multiple cloud services. (Creating)

Syllabus – Cloud Computing

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to Cloud Computing**		**05**
	Understand fundamental concepts, characteristics, components, and deployment models of cloud computing.		
	1.1	Introduction to Cloud Computing, Cloud Computing Architecture, Cloud Characteristics, Cloud Computing Components, Cloud computing benefits and challenges	
	1.2	Cloud Deployment Models: Public, Private, Community, Hybrid	
	1.3	Introduction to popular cloud platforms: AWS, Azure	
	Self-Learning Topics: *Google Cloud Platform (GCP), Multi-cloud and Edge Computing*		
2	**Module 2: Virtualization Concepts and Containerization**		**06**
	Understand the benefits of virtualization and evaluate virtualization and container technologies such as Hypervisors, KVM, Docker, and Kubernetes in cloud environments.		
	2.1	Introduction to Virtualization: Concepts, Benefits, and Design Requirements; Hypervisors and Virtual Machines; Hosted Virtualization	
	2.2	Type-1 and Type-2 Hypervisors: KVM (Kernel-based Virtual Machine) – KVM Architecture, Features and Advantages	
	2.3	Introduction to Kubernetes: Architecture, Pods, Nodes, Deployments, Services, Scaling and Load Balancing, Kubernetes vs Docker, Kubernetes in Cloud Platforms	
	2.4	Containerization Concepts: Containers vs Virtual Machines, Docker Architecture, Images, Containers, and Basic Docker Operations	
	Self-Learning Topics: *Xen Architecture and KVM Virtualization*		
3	**Module 3: Cloud Computing Service Models**		**05**
	Understand and compare IaaS, PaaS, SaaS, and DBaaS with real-life examples.		

	3.1	Cloud Computing: Web Services, Service Oriented Architecture (SOA)	
	3.2	Cloud Computing Service Models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Database as a Service (DaaS) with Concepts and real life examples.	
	3.3	Cloud Computing service model comparison, Responsibilities in each service model	
	Self-Learning Topics: *Applications and use cases of cloud computing (e-commerce, healthcare, IoT)*		
4	**Module 4: Security in Cloud Computing**		**05**
	Learn cloud security fundamentals including IAM, encryption, firewalls, and the AAA model.		
	4.1	Introduction to Cloud Security, Security Challenges and Threats in Cloud Computing, Data Encryption (Data at Rest and Data in Transit), Identity and Access Management (IAM), Cloud Storage Gateways, Cloud Firewalls and Security Groups	
	4.2	AAA (Authentication, Authorization, and Accounting) Model, Single Sign-On (SSO) for Clouds, Authentication Management in Clouds, Authorization Management in Clouds, Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC).	
	Self-Learning Topics: *Security Compliance Standards, Cloud Security Best Practices*		
5	**Module 5: Distributed and Emerging Computing Paradigms**		**05**
	Understand and compare parallel, grid, and quantum computing paradigms and their applications.		
	5.1	Introduction to Computing Paradigms: Evolution, Centralized vs Distributed vs Cloud, High-Performance Computing	
	5.2	Parallel Computing: Types of Parallelism, Shared/Distributed Memory Architectures, Applications	
	5.3	Grid Computing: Architecture, Resource Sharing, Scheduling, Grid vs Cloud, Applications	
	5.4	Quantum Computing: Fundamentals, Qubits, Superposition, Entanglement, Quantum Gates, Applications, Quantum Cloud Platforms	
	Self-Learning Topics: *Quantum Cloud Services (IBM Quantum, Amazon Bracket, Azure Quantum), Edge and Fog Computing*		
6	**Module 6: Cloud Deployment and Resource Management**		**04**
	Deploy applications and manage virtual machines, storage, networking, and monitoring on cloud platforms.		
	6.1	Cloud Resource Provisioning: Creating and Managing VMs(virtual machine) on AWS/Azure, Storage Services, Networking Services, Resource Configuration.	
	6.2	Application Deployment: Web App Deployment, Load Balancing, Auto Scaling, Serverless Computing (AWS Lambda, Azure Functions)	
	Self-Learning Topics: *DevOps Tools in Cloud Computing*		
		TOTAL	**30**

Suggested List of Experiments

Sr. No.	Title of the Experiment
1	**Comparative Case study of Cloud-Based and Non-Cloud-Based Solutions for Real-World Applications.** **Objective:** To understand cloud computing characteristics, components, benefits, challenges, and various cloud deployment models of cloud computing. **Outcome:** Students will be able to explain cloud computing fundamentals and compare Public, Private, Community, and Hybrid cloud deployment models.
2	**Implementation of Virtualization using Hypervisors.** **Objective:** To install and configure Type-1/Type-2 hypervisors and create virtualized environments. **Outcome:** Students will be able to implement virtualization and analyze hypervisor-based resource management.
3	**Creating and Managing Virtual Machines using Infrastructure as a service (IaaS).** **Objective:** To provision, configure, and manage a virtual machine instance on AWS EC2 or Azure Virtual Machines. **Outcome:** Students will be able to deploy and manage computing resources using IaaS services.
4	**Docker Installation and Container Management.** **Objective:** To install Docker and perform container creation, execution, and management operations. **Outcome:** Students will be able to create and manage containerized applications using Docker.
5	**Implementation of Storage as a Service using AWS / Azure Storage.** **Objective:** To create and configure cloud storage resources for storing and managing data. **Outcome:** Students will be able to Create and configure storage services using AWS S3 or Azure Storage. and Manage cloud storage resources and perform data storage operations.
6	**Working with Cloud Storage and Database as a Service (DBaaS).** **Objective:** To create and manage cloud storage and database services on AWS/Azure. **Outcome:** Students will be able to utilize cloud storage and DBaaS services for application deployment.
7	**Deploying a Web Application using Platform as a Service (PaaS).** **Objective:** To deploy a web application on a cloud platform using Platform as a Service (PaaS). **Outcome:** Students will be able to host and manage web applications using cloud platform services.

8	**Configuring Identity and Access Management (IAM).** **Objective:** To create users, roles, and access policies using cloud IAM services. **Outcome:** Students will be able to implement authentication and authorization mechanisms in cloud environments.
9	**Implementing Cloud Security Services using Security as a Service (SaaS).** **Objective:** To configure cloud firewalls, security groups, and data encryption mechanisms. **Outcome:** Students will be able to secure cloud resources using standard cloud security practices.
10	**Monitoring and Managing Cloud Resources.** **Objective:** To monitor cloud resources and application performance using AWS CloudWatch/Azure Monitor. **Outcome:** Students will be able to track resource utilization and manage cloud services efficiently.
12	**Cloud Computing Mini-project** **Objective:** To design and deploy a Web Application Hosted on a Public Cloud Platform incorporating IaaS, PaaS, DBaaS, Storage as a Service and Security as a Service. **Outcome:** Students will be able to design a real-world cloud-based solution and integrate various cloud services. **Note:** AWS/ Azure Tool is Mandatory for implementing all the cloud computing experiments.

Text Books – Cloud Computing

1. Bernard Golden – Amazon Web Services for Dummies, John Wiley & Sons
2. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi – Mastering Cloud Computing, Tata McGraw-Hill Education
3. Gautam Shroff – Enterprise Cloud Computing: Technology, Architecture, Applications, Cambridge University Press, 2010.

Reference Books:

1. Cloud Computing Black Book: Kailash Jayaswal, Jagannath Kallakurchi, Donald J. Houde, Dr. Deven Shah by Kogent Learning Solutions, Publisher: Dreamtech Press.
2. Cloud Computing Concepts Technology and Architecture - Erl second hand book online from Used Books Factory.

Other Resources :

1. NPTEL Swayam Course on Cloud computing By Prof. Soumya Kanti Ghosh https://nptel.ac.in/courses/106/105/106105167/
2. Cloud Computing and Distributed Systems By Prof. Rajiv Misra https://onlinecourses.nptel.ac.in/noc22_cs18/preview
3. Google Cloud Computing Foundation Course https://nptel.ac.in/courses/106/105/106105223

Assessment Methodology – Cloud Computing

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test | Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions are compulsory. Q1 A or B – 10 marks Q2 A or B – 10 marks | Q3 A or B – 10 marks. Each question A and B are based on the same CO. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions are compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus. Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks. Based on performance and satisfactory completion of assigned laboratory work.
PR – Laboratory (25 Marks)	Practical examination based on experiments performed by the students during laboratory sessions.

3b. Cyber Security

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5PCE12	**Cyber Security**	2	1	-	2	1	-	**3**

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	**100**
Tutorial		-	-	-	25	25	-	**50**
Grand Total								**150**

Pre-Requisite Course(s)	25DBFE2VSEC02 - Object Oriented Programming using Python 25CE4PCC03 - Computer Network 25FE2BSC01-Fundamentals of Engineering Mathematics -II

Course Objectives & Module overview

Course Objectives: The Cryptography and System Security (CSS) course aims to provide students with basic understanding of cyber security concepts, cryptographic techniques, and secure communication methods. It helps students apply encryption, analyze security mechanisms like hash functions and digital signatures, and understand network security protocols. The course also focuses on evaluating and designing secure systems using tools such as firewalls and intrusion detection systems to protect against cyber threats.

Module 1: Introduction to Cyber Security

To understand the fundamental concepts of cyber security including security goals (CIA triad), cyber threats, vulnerabilities, risks, and common cyber attacks, and to analyze real-world cyber incidents in terms of CIA triad violations.

Module 2:Operating System Security

To understand and apply operating system security mechanisms such as authentication, access control, user account management, and system protection techniques to safeguard system resources and data from security threats.

Module 3: Number Theory and Basic Cryptography

To understand and apply fundamental number theory concepts such as divisibility, prime numbers, GCD, Euclidean algorithm, modular arithmetic, Fermat's theorem, and Euler's theorem, along with classical cryptographic techniques like substitution and transposition ciphers, to perform basic encryption and decryption for secure communication and problem-solving.

Module 4: Modern Cryptography Concepts

To understand and apply modern cryptography concepts including symmetric and asymmetric encryption, block and stream ciphers, cryptographic hash functions, message authentication codes, digital signatures, and key management techniques to ensure secure communication and data protection.

Module 5: Network Security and Risk Assessment

To understand and apply network security concepts and risk assessment techniques including types of

network attacks, security protocols, firewalls, intrusion detection systems, and risk identification and analysis methods to protect networks and evaluate security risks.

Module 6: Ethical Hacking and System Security

To understand and apply ethical hacking concepts, system vulnerabilities, and system security mechanisms including penetration testing basics, hacking phases, system hardening, patch management, backup and recovery, and security auditing to identify, analyze, and protect systems from cyber attacks.

Overall, the course equips computer science students with fundamental knowledge and practical skills in cyber security, cryptography, and system security to analyze threats, apply security mechanisms, and design secure computing systems for protecting data and networks.

Course Outcomes – Cyber Security

CO	Course Outcome
CO1	Recall the fundamental principles of Cyber Security and explain the CIA Triad, threats, vulnerabilities, and common cyber attacks.
CO2	Discuss the role of authentication methods, access control mechanisms, and common security threats in Operating System Security.
CO3	Apply number theory and classical cryptographic techniques to perform basic encryption and decryption operations.
CO4	Analyze symmetric, asymmetric, and hash-based cryptographic systems for secure communication and data integrity.
CO5	Evaluate network security mechanisms and risk assessment techniques to identify and reduce security risks in systems.
CO6	Design basic security solutions using ethical hacking concepts and system security mechanisms to protect systems from cyber threats.

Syllabus – Cyber Security

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to Cyber Security**		**03**
	Understand fundamental concepts of cyber security including the CIA triad, cyber threats, vulnerabilities, and common attacks.		
	1.1	Fundamentals of Cyber Security, Need and Importance, Security Goals (CIA), Cyber Threats, Vulnerabilities and Risks, Types of Cyber Attacks (Malware, Phishing, Social Engineering, DoS/DDoS), Role of Cryptography	
	Self-Learning Topics: *Case Study: Analyze a real cyber attack and identify the breach in the CIA triad*		
2	**Module 2: Operating System Security**		**04**
	Identify common OS threats and recall authentication and access control mechanisms.		
	2.1	OS Security Fundamentals: Threats and Vulnerabilities, Security Principles; Access Control: User Authentication, MFA, ACL, File/Directory Permissions	
	2.2	Malware and System Protection: Viruses, Worms, Trojans, Ransomware, Spyware, Detection and Prevention	

	Self-Learning Topics: *Comparative study of Password-based Authentication vs Multi-Factor Authentication (MFA)*		
3	**Module 3: Number Theory and Basic Cryptography**		**08**
	Apply number theory concepts and classical cryptographic techniques for encrypting and decrypting messages.		
	3.1	Number Theory Foundations: Divisibility, Prime Numbers, GCD, Euclidean Algorithm, Modular Arithmetic, Fermat's Theorem, Euler's Theorem	
	3.2	Classical Cryptography: Symmetric Cipher Model, Caesar Cipher, Monoalphabetic, Playfair, Vigenere, Rail Fence, and Columnar Transposition Ciphers	
	Self-Learning Topics: *Encrypt a sentence using two classical ciphers and compare results*		
4	**Module 4: Modern Cryptography Concepts**		**10**
	Differentiate symmetric and asymmetric cryptography and analyze digital signatures and authentication protocols.		
	4.1	Symmetric and Asymmetric Cryptography: DES, AES, RSA; Block Cipher vs Stream Cipher	
	4.2	Hash Functions and Message Digest: MD5, SHA family, applications	
	4.3	Digital Signature: Concept, generation and verification; Key Management: KDC, Diffie-Hellman, Kerberos	
	Self-Learning Topics: *Study of cryptography in online banking, email security, and secure websites*		
5	**Module 5: Network Security and Risk Assessment**		**02**
	Evaluate the impact of network attacks and assess organizational risks with mitigation strategies.		
	5.1	Network Security Fundamentals: TCP/IP vulnerabilities, Packet Sniffing, ARP Spoofing, IP Spoofing, Port Scanning, DoS and DDoS	
	5.2	Security Mechanisms: Firewalls, IDS, IPS	
	5.3	Risk Assessment: Risk, Threat and Vulnerability concepts, Qualitative and Quantitative Risk Assessment, Mitigation Strategies	
	Self-Learning Topics: *Study a recent cyber attack and evaluate security measures that could reduce its impact*		
6	**Module 6: Ethical Hacking and System Security**		**03**
	Design basic security approaches using ethical hacking and system security mechanisms.		
	6.1	Ethical Hacking Basics, Types of Hackers, Phases of Ethical Hacking, Vulnerability Assessment, Penetration Testing, Security Hardening and Monitoring	
		TOTAL	**30**

Suggested List of Experiments

S. No.	List of Experiments
1	**Module 1: Introduction to Cyber Security** **Objectives:** ● To analyze the CIA triad (Confidentiality, Integrity, and Availability) in cyber security through real-world scenarios. **Outcomes:** ● Students will be able to identify and analyze the impact of cyber attacks on the CIA triad in given situations. **Experiment no 1 : CIA Triad Analysis**
2	**Module 2: Operating System Security** ● **Objectives:** To perform and observe basic operating system security mechanisms such as authentication, access control (file permissions), and malware protection using system tools. ● **Outcomes:** Students will be able to demonstrate and identify operating system security features including user authentication, file and directory permissions, and basic malware detection techniques. **Experiment 2 : To observe and verify basic operating system security mechanisms such as authentication, access control, and malware protection features in a system.**
3	**Module 3:Number Theory and Basic Cryptography** ● **Objectives:** To demonstrate the combined use of substitution and transposition techniques for securing messages. ● **Outcomes:** **S**tudents will be able to encrypt and decrypt messages using a product cipher. **Experiment 3: Design and Implementation of a Product Cipher using Substitution and Transposition Ciphers.**
4	Module 4: **Modern Cryptography Concepts** ● **Objectives:** To understand the working of public key cryptography through RSA. ● **Outcomes:**

	Students will be able to perform RSA-based encryption and decryption. **Experiment 4:Implementation and analysis of RSA crypto system.**
5	**Module 4: Modern Cryptography Concepts** ● **Objectives:** To demonstrate secure key establishment between communicating parties. ● **Outcomes:** Students will be able to establish a shared secret key using the Diffie-Hellman algorithm. **Experiment 5:Implementation of Diffie Hellman Key exchange algorithm.**
6	**Module 5: Network Security and Risk Assessment** ● **Objectives:** To investigate network traffic and packet analysis techniques. ● **Outcomes:** Students will be able to capture, filter, and analyze network packets using Wireshark. **Experiment 6: Study of packet sniffer tools: wireshark,:** **1. Download and install wireshark and capture icmp, tcp, and http packets in promiscuous mode.** **2. Explore how the packets can be traced based on different filters.**
7	**Module 5: Network Security and Risk Assessment** **Objectives:** To explore various network scanning and host discovery techniques. **Outcomes:** Students will be able to identify open ports and gather network information using Nmap. **Experiment no 7 :Download and install nmap. Use it with different options to scan open ports, perform OS fingerprinting, do a ping scan, tcp port scan, udp port scan, xmas scan**
8	**Module 2: Operating System Security** **Objectives:** To identify and analyze the impact of software vulnerabilities through buffer overflow attacks. **Outcomes:** Students will be able to identify and analyze buffer overflow vulnerabilities.

	Experiment 8: Simulate buffer overflow attack using Ollydbg, Splint, Cpp check
9	**Module 5 : Network Security and Risk Assessment** **Objectives:** To evaluate system security risks by implementing a Python program using a predefined dataset and calculating risk scores based on likelihood and impact values. **Outcomes:** Students will be able to evaluate and classify security risks in a system using a Python-based predefined dataset approach without manual input. **Experiment 9 :Risk Calculation and Assessment Using Python with Predefined Dataset**
10	**Module 6 : Ethical Hacking and System Security** **Objectives:** To design a basic ethical hacking framework for identifying system vulnerabilities and suggesting appropriate security measures. . **Outcomes:** Students will be able to design and present a simple ethical hacking model including phases of hacking and corresponding security countermeasures for a given system. **Experiment 10 : Design of Basic Ethical Hacking Framework for a System**
Implementation of basic cryptographic algorithms, number theory concepts, and system security mechanisms using suitable programming approaches to reinforce theoretical concepts of cyber security and secure communication.	

Text Books – Cyber Security

1. William Stallings – Cryptography and Network Security: Principles and Practice, 6th Edition, Pearson Education, 2013
2. Behrouz A. Forouzan – Cryptography and Network Security, Tata McGraw-Hill
3. Behrouz A. Forouzan & Debdeep Mukhopadhyay, “Cryptography and Network Security” 3rd Edition, McGraw Hill

Reference Books – Cyber Security

1. Atul Kahate, “Cryptography and Network Security”, Tata McGraw-Hill Education, 2003.
2. Bruce Schneier, “Applied Cryptography, Protocols Algorithms and Source Code in C”,Second Edition, Wiley.
3. Eric Cole, “Network Security Bible”, Second Edition, Wiley, 2011.

Useful Links

- https://nptel.ac.in/courses/106106222?utm_source=chatgpt.com
- https://owasp.org/www-project-top-ten/?utm_source=chatgpt.com
- https://cryptii.com/?utm_source=chatgpt.com
- https://www.cloudflare.com/learning/security/?utm_source=chatgpt.com

Assessment Methodology – Cyber Security

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test | Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions compulsory. Q1 A or B – 10 marks Q2 A or B – 10 marks | Q3 A or B – 10 marks. Each question A and B based on same CO. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus. Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks. Based on performance and satisfactory completion of assigned laboratory work.
OR– Laboratory (25 Marks)	Oral examination based on experiments performed by the students during laboratory sessions.

3c. Statistics for Machine Learning and Data Science

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5PCE013	**Statistics for Machine Learning and Data Science**	2	2	-	2	1	-	**3**

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	**100**
Lab		-	-	-	25	25	-	**50**
Grand Total								**150**

Pre-Requisite Course(s)	25FE1BSE01 – Engineering Mathematics 01 25FE2VSEC02 – Object Oriented Programming Using Python

Course Objectives & Module overview

Statistics for Machine Learning and Data Science provides a comprehensive foundation in statistical methods and their applications in Data Science and Machine Learning. It introduces descriptive statistics, probability theory, sampling techniques, regression analysis, statistical estimation, and hypothesis testing. Students will learn how to collect, analyze, interpret, and draw inferences from data using statistical tools and techniques. The course emphasizes practical applications of statistics in predictive modeling, machine learning, and data-driven decision-making. By the end of the course, students will be able to apply statistical concepts to real-world datasets, evaluate machine learning models, and make informed decisions based on statistical evidence.

Course Objectives:

1.To introduce fundamental concepts of statistics, probability, and data analysis.
2.To develop understanding of statistical measures, probability laws, and inference techniques.
3.To apply statistical methods and regression models for data analysis.
4.To analyze relationships among variables using statistical and probabilistic techniques.
5.To evaluate statistical models and inference methods for decision making
6.To develop statistical solutions for Machine Learning and Data Science applications.

Module 1: This module introduces the fundamental concepts of statistics and their significance in data science and machine learning. Students will learn methods for organizing, summarizing, and visualizing data using statistical measures and graphical techniques. The module also covers measures of central tendency, dispersion, skewness, kurtosis, correlation, covariance, and distance metrics used in exploratory data analysis and feature selection.

Module 2: This module covers the foundations of probability theory and random variables. Students will learn probability laws, conditional probability, Bayes' theorem, Bayesian inference, and probability

distributions. The module introduces discrete and continuous random variables, statistical independence, joint and conditional distributions, and commonly used probability distributions in machine learning applications.

Module 3: This module focuses on data collection methodologies and sampling techniques used in statistical studies. Students will understand population and sample concepts, census and sample surveys, probability and non-probability sampling methods, bootstrap sampling, sampling errors, and sampling distributions. The Law of Large Numbers and Central Limit Theorem are introduced as the foundation for statistical inference.

Module 4: This module introduces regression analysis as a tool for modeling relationships among variables and making predictions. Students will learn simple and multiple linear regression, model assumptions, significance testing, and model evaluation metrics such as R^2, MAE, MAPE, MSE, and RMSE. The module emphasizes the application of regression techniques in predictive analytics and machine learning.

Module 5: This module introduces statistical inference through estimation theory. Students will learn point estimation, properties of estimators, Method of Moments, Maximum Likelihood Estimation (MLE), confidence intervals, sample size determination, and applications of confidence intervals in model evaluation. The module highlights the role of estimation techniques in machine learning and statistical modeling.

Module 6: This module covers the principles of hypothesis testing and statistical decision-making. Students will study standard statistical distributions, formulation of null and alternative hypotheses, Type I and Type II errors, Neyman-Pearson lemma, Most Powerful (MP) tests, and Uniformly Most Powerful (UMP) tests. The module emphasizes the use of hypothesis testing techniques in machine learning experiments, model comparison, and data-driven decision-making.

Course Outcomes – Statistics for ML & DS

CO	Course Outcome
CO1	**Recall** the fundamental concepts of statistics, probability, data collection methods, sampling techniques, and statistical distributions used in data science and machine learning (Remembering)
CO2	**Explain** descriptive statistical measures, probability concepts, random variables, sampling distributions, and regression principles for data analysis. (Understanding)
CO3	**Apply** probability laws, sampling methods, estimation techniques, confidence intervals, and regression models to solve data science and machine learning problems. (Applying)
CO4	**Analyze** relationships among variables using correlation analysis, probability models, simple and multiple regression techniques, and statistical tests. (Analyzing)
CO5	**Evaluate** estimators, confidence intervals, regression models, and hypothesis testing procedures for statistical inference and decision-making. **(**Evaluating)
CO6	**Design** and develop data-driven solutions by integrating statistical analysis, predictive modeling, estimation, and hypothesis testing techniques for real-world machine learning applications. (Creating)

Syllabus – Statistics for ML & DS

Module No.	Unit No.	Topics	Hours
1	**Module 1: Descriptive Statistics and Data Representation**		**03**
	After completing this module, students will be able to recall the fundamental concepts of statistics, data classification, descriptive statistical measures, and data visualization techniques used in data analysis.		
	1.1	**Functions, importance, uses and limitations of Statistics.**	
	1.2	**Data Classification and Visualization:** Statistical data– classification, tabulation, descriptive statistics:central tendency, dispersion, skewness, kurtosis. , diagrammatic and graphical representation of data;histograms, boxplots, scatter plots, pair plots.	
	1.3	**Statistical Measures:** Expectation, Variance, Covariance, correlation, confounding, correlation matrices and their properties.Distance measures: Euclidean, Manhattan, Cosine,Introduction to Feature Selection using Correlation ,	
	Self-Learning Topics: Exploratory Data Analysis (EDA), Outlier Detection Methods, Data Preprocessing Concepts		
2	**Module 2: Probability and Random Variables**		**05**
	After completing this module, students will be able to explain probability concepts, random variables, statistical independence, and probability distributions for modeling uncertainty in data-driven applications.		
	2.1	**Fundamentals of Probability:** Experiment, Sample Space and Events, Types of Events, Simple and Compound Events, Mutually Exclusive Events, Exhaustive Events, Independent and Dependent Events, Conditional Probability, Baye's Theorem, Bayesian Inference (Prior, Posterior, Likelihood) ,Applications of Probability in Data Science and Machine Learning.	
	2.2	**Random Variables and Statistical Independence:** Concept of Random Variable, Discrete and Continuous Random Variables, Statistical Independence.	
	2.3	**Multivariate Probability Distributions:** Joint, Marginal and Conditional Distributions of Two-Dimensional Random Variables.	
	2.4	**Probability Distributions:**Probability Mass Function (PMF),Probability Density Function (PDF),Cumulative Distribution Function (CDF),Binomial distribution,Poisson distribution,Uniform distribution,Normal distribution.	
	Self-Learning Topics: Introduction to Information Theory (Entropy), Probabilistic Graphical Models, Applications of Bayesian Networks.		
3	**Module 3: Data Collection & Sampling Methods**		**04**
	After completing this module, students will be able to apply appropriate data collection methods, sampling techniques, and sampling distribution concepts for statistical analysis and inference.		

	3.1	**Data Sources and Collection Techniques:** Primary and Secondary Data, Sources of Data, Methods of Data Collection.	
	3.2	**Sampling Concepts and Methods:** Population and Sample,Census & Sample methods –Methods of sampling, Probability Sampling and Non-Probability Sampling Techniques: Simple Random, Stratified, Cluster, Bootstrap Sampling Techniques and Applications ,Sampling and Non-Sampling Errors.	
	3.3	**Theory of Sampling Distributions:** Sampling Distribution of Means, Law of Large Numbers and Central Limit Theorem.	
	Self-Learning Topics: Missing Data Mechanisms, Data Bias and Fairness, Resampling Techniques		
4	**Module 4: Regression Analysis and Predictive Modeling**		**10**
	After completing this module, students will be able to analyze relationships among variables and evaluate regression models using statistical and predictive performance measures.		
	4.1	Correlation and Dependence Analysis.	
	4.2	**Fundamentals of Linear Regression**: Mathematical and Statistical Equations, Simple Linear Regression, Meaning of Intercept and Slope.	
	4.3	**Regression Model Evaluation:** Error Term, Assumptions of Regression, Measures of Model Fit: R^2, MAE, MAPE, MSE, RMSE, Adjusted R^2.	
	4.4	**Multiple Regression and Significance Testing:** Multiple Linear Regression Model, Partial Regression Coefficients, Testing Significance overall significance of Overall fit of the model, Testing for Individual Regression Coefficients	
	Self-Learning Topics: Introduction to Logistic Regression, Overfitting and Underfitting		
5	**Module 5: Statistical Inference and Estimation**		**04**
	After completing this module, students will be able to evaluate estimators, confidence intervals, and statistical inference methods for estimating population parameters and assessing model reliability.		
	5.1	Random sample -Parametric point estimation unbiasedness and consistence-Method of Moments and Maximum Likelihood Estimation (MLE).	
	5.2	**Sample Size Determination and Model Evaluation:** Confidence Intervals for Variance, Sample Size Determination and Applications of Confidence Intervals in Model Performance Evaluation.	
	Self-Learning Topics:Resampling Methods, Introduction to Statistical Learning Theory.		
6	**Module 6: Tests of hypotheses**		**04**
	After completing this module, students will be able to design and implement statistical decision-making frameworks by applying hypothesis testing techniques and statistical distributions to solve real-world data science and machine learning problems.		
	6.1	**Standard Statistical Distributions:** Population Distribution, Z-Distribution, Student's t-Distribution, Chi-Square Distribution and F-Distribution.	
	6.2	Null and Alternative hypotheses. Types of errors. Neyman-Pearson lemma MP and UMP tests.	
	Self-Learning Topics: Non-Parametric Tests, Multiple Hypothesis Testing,		

TOTAL	**30**

Laboratory Experiments – Statistics for ML & DS (Minimum 10)

No.	Title and Details
1	**Aim: To perform descriptive statistical analysis and visualize data using graphical techniques.** **Objective: Compute descriptive statistical measures (central tendency, dispersion, skewness, kurtosis) and visualize data using histograms, boxplots, scatter plots, and pair plots.** **Outcome: Students will be able to summarize datasets and interpret statistical characteristics using visualization tools.**
2	**Aim: To analyze relationships among variables using statistical and distance measures.** **Objective: Quantify relationships using covariance and correlation; compute distance metrics for similarity analysis.** **Outcome: Students will be able to measure dependence and similarity among variables in datasets.**
3	**Aim: To understand and implement basic probability concepts and event relationships.** **Objective: Apply probability rules, conditional probability, and Bayes' theorem to decision-making scenarios.** **Outcome: Students will be able to solve probability problems and interpret uncertainty in data-driven applications.**
4	**Aim: To analyze multivariate probability distributions.** **Objective: Compute joint, marginal, and conditional probabilities from multivariate datasets.** **Outcome: Students will be able to model probabilistic relationships among multiple variables.**
5	**Aim: To generate and analyze discrete and continuous probability distributions.** **Objective: Study the behavior of random variables through PMF, PDF, and CDF visualizations.** **Outcome: Students will be able to identify appropriate probability models for different data scenarios.**
6	**Aim: To study different sampling methods and evaluate sampling errors.** **Objective: Implement various sampling methods and compare sample statistics with population characteristics.** **Outcome: Students will be able to select suitable sampling techniques for data collection and analysis.**
7	**Aim:To demonstrate sampling distributions and the Central Limit Theorem through simulation.** **Objective:To study sampling distributions through repeated random sampling experiments and verify the Law of Large Numbers and Central Limit Theorem using simulations.** **Outcome:Students will be able to explain the foundation of statistical inference using simulations.**
8	**Aim:To develop a predictive model using simple linear regression.** **Objective:To develop a simple linear regression model for analyzing relationships between variables and interpret regression coefficients to evaluate prediction performance.** **Outcome:Students will be able to predict outcomes and interpret linear relationships between variables.**

9	**Aim:To build and evaluate multiple linear regression models.** **Objective:To construct multiple linear regression models using multiple predictor variables and assess model performance using statistical and error-based evaluation metrics.** **Outcome:Students will be able to assess model performance and identify influential predictors.**
10	**Aim:To estimate population parameters and construct confidence intervals.** **Objective:To estimate population parameters using point and interval estimation techniques and construct confidence intervals and apply maximum likelihood estimation methods. Outcome:Students will be able to evaluate uncertainty associated with parameter estimates.**
11	**Aim:To study hypothesis testing concepts and analyze Type I and Type II errors using statistical distributions.** **Objective: To formulate null and alternative hypotheses for statistical decision-making problems and determine critical regions and decision rules using Z, t, Chi-Square, and F distributions.** **Outcome:Students will be able to formulate statistical hypotheses, interpret decision rules, and evaluate the effects of Type I and Type II errors in statistical inference.**

Text Books – Statistics for ML & DS

1. Agarwal, B.L. (2006) – Basic Statistics, Wiley Eastern Ltd., New Delhi
2. Gupta, S. P. (2011) – Statistical Methods, Sultan Chand & Sons, New Delhi
3. Hogg, R.V. and Craig, A.T. (2006) – An Introduction to Mathematical Statistics, Amerind Publications
4. Miller & Freund – Probability and Statistics for Engineers, 9th Edition

Reference Books – Statistics for ML & DS

1. Arora, P.N., Sumeet Arora, S. Arora (2007) – Comprehensive Statistical Methods, Sultan Chand, New Delhi
2. Montgomery, D.C., Peck, E.A., & Vining, G.G. (2003) – Introduction to Linear Regression Analysis, John Wiley and Sons
3. Mood, A.M., Graybill, F.A., and Boes, D.C. (1985) – Introduction to the Theory of Statistics, McGraw-Hill

Useful Links – Statistics for ML & DS

1. https://onlinecourses-archive.nptel.ac.in/noc19_ma08/preview
2. https://onlinecourses-archive.nptel.ac.in/noc18_ma22/preview

Assessment Methodology – Statistics for ML & DS

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test | Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions compulsory. Q1 A or B – 10 marks

	Q2 A or B – 10 marks | Q3 A or B – 10 marks. Each question A and B based on same CO. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus. Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks. Based on performance and satisfactory completion of assigned laboratory work.
OR – Laboratory (25 Marks)	Oral examination based on experiments performed by the students during laboratory sessions.

4. Android Application Development

Course Code: 25CE5MDM01 | Category: MDM 02

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5MDM01	**Android Application Development**	3	2	-	3	1	-	**4**

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		20	30	50	-	-	-	**100**
Lab		-	-	-	25	-	-	**25**
Grand Total								**125**

Course Objectives & Module overview

Course and Module Overview:

This course is designed to introduce Mechanical and Electronics Engineering students to the fundamentals of Android application development. Offered by the Department of Computer Engineering, it provides a structured, beginner-friendly pathway to understanding the mobile application ecosystem, development tools, and core programming concepts needed to build functional Android apps.

The course focuses on hands-on learning through six progressively structured modules, starting from environment setup and basic UI design, to data management, background services, and networked applications. Students will gain practical experience through a minimum of twelve laboratory sessions covering real-world development scenarios.

The curriculum is kept intentionally broad to ensure accessibility for students from non-software backgrounds, while still delivering industry-relevant skills. On completing this course, students will be capable of independently developing, testing, and deploying simple Android applications, and will have a strong foundation for further exploration of mobile development in their respective engineering domains.

Module 1 – Introduction to Android

This module builds the foundational knowledge required to begin Android development. Students will explore the history and growth of the Android ecosystem, understand the overall architecture of Android applications, and learn to configure the development environment including Android Studio and the SDK. The module covers project structure, the Android Manifest file, running apps on virtual and physical devices, and the Activity lifecycle. The SOLID design principles are also introduced to encourage good software design habits from the outset.

Module 2 – Android UI Fundamentals

This module focuses on designing and building effective user interfaces using Android's XML-based layout system. Students will work with the most commonly used UI components including TextView, EditText, Button, and ImageView, and will learn to structure screens using LinearLayout, RelativeLayout, and ConstraintLayout. Topics also include handling screen orientation changes, using the action bar, displaying Toast notifications, and applying basic UI design principles to build clean and intuitive interfaces.

Module 3 – Activities, Intents & Fragments

This module explores the structural building blocks of Android applications. Students will develop a thorough understanding of the Activity lifecycle and learn to navigate between screens using explicit and implicit Intents. Intent filters and data passing between activities are covered in detail. The module also introduces Fragments, their lifecycle, management, and use in building flexible multi-screen applications, giving students the skills needed to construct well-structured, navigable apps.

Module 4 – Data Storage

This module equips students with the techniques needed to persist data within Android applications. Topics include SharedPreferences for storing small key-value data, reading and writing to internal device storage, and creating and managing a SQLite relational database through direct SQL operations and the higher-level Room Persistence Library. Students will implement basic Create, Read, Update, and Delete (CRUD) operations and understand when to apply each storage mechanism based on the nature and volume of the data.

Module 5 – Android Services & Notifications

This module introduces the mechanisms Android provides for running operations in the background and communicating with the user. Students will learn about the Service lifecycle, implement Broadcast Receivers to respond to system and app-level events, and use NotificationManager to deliver alerts and updates. The module also covers scheduling tasks with AlarmManager, sending emails via Intent, and managing user alerts, skills essential for building responsive and professionally polished applications.

Module 6 – Networking & Multimedia

This module introduces students to connecting Android applications to the internet and handling rich media content. Topics include making HTTP requests to web servers, parsing JSON responses, and using the Retrofit library for efficient API consumption. Students will also explore Android's Media APIs to integrate audio and video playback into their apps. The module concludes with an overview of the AndroidX libraries and an introduction to RESTful web service integration, preparing students for building data-driven, connected applications.

Course Outcomes

CO	Course Outcome
CO1	Recall the Android ecosystem, architecture, and development environment. (Remembering)
CO2	Understand user interfaces using XML layouts and standard UI components such as TextView, EditText, Button, and ImageView. (Understanding)
CO3	Apply Activity lifecycle concepts, use explicit and implicit Intents, to develop multi-screen applications using Fragments. (Applying)
CO4	Illustrate data storage mechanisms using SharedPreferences, internal file storage, and SQLite database for data persistence. (Analyzing)
CO5	Evaluate Android services, broadcast receivers, notification services, and alarm management in application development. (Evaluating)
CO6	Develop Android applications by integrating networking features, consuming web services, parsing JSON data, and utilizing multimedia APIs for audio and video playback. (Creating)

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to Android**		**08**
	Understand Android ecosystem and evolution, explain Android architecture and components, set up the development environment.		
	1.1	Android Ecosystem Overview: Devices, OS layers, Play Store ecosystem, Open-source nature	
	1.2	History & Versions: Evolution timeline, major releases, feature improvements, API levels	
	1.3	Development Setup: Android Studio installation, SDK configuration, Emulator setup, Running Hello World	
	1.4	Project Structure & Architecture: Manifest file, Project folders, AAC components, Activity lifecycle basics	
	Self-Learning Topics: *Explore latest Android version features; Compare Android with*		

	other mobile OS		
2	**Module 2: Android UI Fundamentals**		**08**
	Design UI using XML layouts, use common UI components, handle screen orientation, implement user interaction feedback.		
	2.1	XML Layout Design: Layout structure, View hierarchy, Attributes, UI rendering	
	2.2	UI Components: TextView, EditText, Button, ImageView properties and usage	
	2.3	Layout Types: LinearLayout, RelativeLayout, ConstraintLayout, Use cases	
	2.4	UI Interaction: Orientation changes, Action bar, Toast messages, Basic notifications	
	Self-Learning Topics: *Design sample mobile app screens; Create responsive layouts*		
3	**Module 3: Activities, Intents & Fragments**		**08**
	Understand activity lifecycle, use intents for communication, implement fragments, design multi-screen apps.		
	3.1	Activity Lifecycle: States, transitions, callbacks, lifecycle management	
	3.2	Intents: Explicit, implicit, intent filters, data passing	
	3.3	Application Flow: Multi-screen navigation, back stack, UI transitions	
	3.4	Fragment Communication: Passing data between fragments, fragment-to-activity interaction, ViewModel usage basics	
	Self-Learning Topics: *Implement fragment-based UI*		
4	**Module 4: Data Storage**		**08**
	Store and retrieve data using SharedPreferences, internal storage, SQLite, and Room library.		
	4.1	SharedPreferences: Key-value storage, reading/writing data, use cases	
	4.2	Internal Storage: File handling, read/write operations, security aspects	
	4.3	SQLite Database: Database creation, CRUD operations, queries	
	4.4	Room Library: ORM concepts, advantages, entity and DAO basics	
	Self-Learning Topics: *Develop small data storage app; Compare SQLite and Room*		
5	**Module 5: Android Services & Notifications**		**08**
	Understand background processing, use services and receivers, implement notifications and handle user alerts.		
	5.1	Android Services: Types, lifecycle, background execution, use cases	
	5.2	Broadcast Receivers: Event handling, system broadcasts, custom broadcasts	
	5.3	Notifications: Notification manager, alerts, alarm services, user interaction	
	5.4	Intent-based Communication: Sending emails, implicit intents, external apps interaction	
	Self-Learning Topics: *Explore background task handling*		
6	**Module 6: Networking & Multimedia Basics**		**05**

	Consume web services, handle JSON data, use networking libraries and multimedia APIs.		
	6.1	Networking Basics: HTTP methods, API calls, connectivity handling	
	6.2	JSON Processing: Parsing techniques, data mapping, error handling	
	6.3	Retrofit Library: Setup, API integration, advantages	
	6.4	Multimedia: Audio/video playback, media APIs, AndroidX libraries overview	
	Self-Learning Topics: *Create API-based app; Integrate media playback*		
		TOTAL	**45**

List of Assignments

- Develop a basic Android application demonstrating core setup and execution.
- Create a fragment-based user interface for dynamic screen management.
- Implement navigation between multiple screens using activities and intents.
- Case study on modern networking approaches and API integration in Android applications.
- Study of architecture of Android and explain its core components.

Suggested List of Activities

No.	Title and Details
1	**AIM: Set Up Android Studio and Create a Basic Android Application** **Objective: Install Android Studio and configure emulator; Create and run a Hello World application; Understand project structure and Manifest file; View log statements using Logcat.** **Outcome: Students will be able to set up Android environment and execute a basic application successfully.**
2	**AIM: Develop a Basic Interactive UI Application Using Layouts and Button Events** **Objective: Display Toast message on button click; Explore LinearLayout, RelativeLayout, ConstraintLayout; Handle basic UI interactions.** **Outcome: Students will be able to create interactive UI applications with event handling.**
3	**AIM: Design a User Input Form and Transfer Data Between Activities** **Objective: Create form with multiple input fields; Collect user details (name, email, etc.); Pass data using Explicit Intent.** **Outcome: Students will be able to design forms and implement activity-to-activity communication.**
4	**AIM: Implement Implicit Intents and Intent Filters in Android** **Objective: Open web pages using implicit intents; Launch Google Maps using location intent; Implement intent filters for activity access.** **Outcome: Students will be able to use system-level apps through implicit intents.**
5	**AIM: Implement Various Input Controls and Picker Dialogs** **Objective: Use Spinner for dropdown selection; Implement AlertDialog with actions; Use DatePicker and TimePicker dialogs.** **Outcome: Students will be able to integrate advanced input components in Android apps.**
6	**AIM: Implement Menus, Selection Controls, and Navigation** **Objective: Create Options/Overflow menu; Handle menu item selection; Use Radio Buttons; Implement Up navigation in App Bar.**

	Outcome: Students will be able to design structured navigation and selection interfaces.
7	**AIM: Apply Themes, Styles, and Custom Drawable Resources in Android** **Objective: Create custom themes and styles; Apply reusable UI styling; Use drawable resources in UI design.** **Outcome: Students will be able to enhance UI consistency using themes and styles.**
8	**AIM: Implement User Preferences Using SharedPreferences and Settings Screen** **Objective: Create Settings screen using PreferenceFragmentCompat; Store and retrieve user preferences; Apply preferences at runtime.** **Outcome: Students will be able to manage user settings and persistent preferences.**
9	**AIM: Develop a CRUD-Based Notes Application Using SQLite Database** **Objective: Perform insert, update, delete, and read operations; Store notes in SQLite database; Display data using RecyclerView.** **Outcome: Students will be able to implement database-driven applications with CRUD operations.**
10	**AIM: Perform Background Processing Using AsyncTask or WorkManager** **Objective: Create background tasks; Update UI after task completion; Understand threading concepts.** **Outcome: Students will be able to execute background operations in Android applications.**

Text Books

1. Wei Meng Lee – Beginning Android 4 Application Development, Wrox Publication

Reference Books

2. Joseph Annuzzi, Lauren Darcey, Shane Conder – Advanced Android Application Development, 4th Edition, Pearson
3. John Horton – Android Programming for Beginners, 3rd Edition, Packt Publication
4. Pradeep Kothari – Android Application Development Black Book, DreamTech

Assessment Methodology

Type of Assessment	Assessment Tools / Marks Distribution
CA – Theory (20 Marks)	Certification: NPTEL (20 Marks) OR Any two Pedagogies (10 marks each): MCQ/Class Test Case study/Assignment GATE based Assignment Certification: Udemy/Coursera Open Book Test Working model/simulation
MSE – Theory (30 Marks)	All Questions compulsory. Q1 A or B – 10 marks Q2 A or B – 10 marksQ3 A or B – 10 marks. MSE based on 50% syllabus. Time: 90 minutes. Total Marks: 30
ESE – Theory (50 Marks)	All Questions compulsory. Q1-Q5, each A or B – 10 marks. ESE: 30% MSE syllabus + 70% post-MSE syllabus.

	Time: 120 minutes. Total Marks: 50
TW – Laboratory (25 Marks)	Active Participation (Lab) = 5 marks Laboratory Report = 10 marks Laboratory Performance = 10 marks.

5. Open Elective – OE 01

Course Code: 25CE5OE01 | Category: OE

Field	Details
Course	Students choose one Open Elective from those offered by other departments or as approved by the university. The detailed syllabus is provided separately by the respective offering department.
Credits	As per the offering department / university norms
Teaching Scheme	As per the offering department

The Open Elective enables students to acquire knowledge in areas outside their core engineering discipline, broadening their technical and interdisciplinary competence in accordance with NEP 2020 guidelines.

6. Professional Communication Skills

Course Code: 25CE5AEC02 | Category: AEC (Ability Enhancement Course)

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5AEC02	Professional Communication Skills	2+2	-	-	2	-	-	2

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		-	-	-	50	-	-	50

Pre-Requisite Course(s)	25FEAEC01 – Effective Communication Skills

Course Objectives and Module Overview

Course Objectives:

This course is designed to develop professional communication competencies required in academic, technical, corporate, and global workplace environments. The course focuses on enhancing verbal, written, interpersonal, digital, and presentation communication skills essential for engineering graduates and professionals. Students will develop the ability to communicate effectively in professional settings through technical writing, presentations, group discussions, workplace interactions, and digital communication platforms.

The course also emphasizes leadership communication, critical thinking, emotional intelligence, corporate etiquette, teamwork, and interview readiness. Practical activities such as mock interviews, presentations, group discussions, report writing, and roleplay simulations provide experiential learning opportunities to prepare students for placements, internships, entrepreneurship, and professional careers.

The course integrates modern communication practices including AI-assisted communication tools, digital collaboration platforms, professional networking, and global workplace communication standards to ensure industry relevance and employability readiness.

Module 1 This module develops advanced professional communication competencies required in workplace and corporate environments. It focuses on persuasive communication, negotiation skills, emotional intelligence, workplace ethics, and cross-cultural communication. Students will learn to communicate strategically, handle professional interactions effectively, and resolve workplace conflicts through appropriate communication practices.

Module 2 This module emphasizes professional writing skills required for technical and corporate communication. Students will learn technical report writing, proposal drafting, executive summaries, research documentation, professional emails, meeting documentation, and AI-assisted writing practices used in industry and academic environments

Module 3 This module develops professional speaking and interpersonal communication abilities. It focuses on presentations, group discussions, panel discussions, interview communication, storytelling techniques, and corporate meeting simulations. Students will strengthen public speaking confidence, teamwork communication, and leadership interaction skills.

Module 4 This module introduces modern digital communication platforms and AI-based communication tools used in professional environments. Students will learn professional networking, digital etiquette, prompt engineering basics, AI-assisted content generation, online collaboration, and personal branding strategies for digital workplace communication.

Module 5 This module prepares students for internships, placements, and professional careers through

resume building, portfolio development, interview preparation, corporate professionalism, and workplace communication practices. Students will gain confidence in handling HR, technical, behavioral, and stress interviews effectively.

Module 6 This module focuses on leadership communication, innovation pitching, entrepreneurship communication, design thinking, crisis communication, and multicultural workplace interaction. Students will develop the communication competencies required for leadership roles, global collaboration, and future workplace environments.

Course Outcomes

CO	Course Outcome
CO1	Define the fundamentals of professional, technical, digital, and workplace communication. (Remembering)
CO2	Explain communication models, professional ethics, corporate communication practices, and workplace interaction strategies. (Understanding)
CO3	Apply verbal, written, presentation, and digital communication skills in academic, technical, and professional contexts. (Applying)
CO4	Analyze workplace communication situations, group discussions, negotiation scenarios, and professional challenges to identify suitable communication strategies. (Analyzing)
CO5	Evaluate leadership communication, ethical communication practices, and cross-cultural workplace interactions in professional environments. (Evaluating)
CO6	Create technical reports, business documents, presentations, resumes, portfolios, and AI-assisted communication content for industry readiness. (Creating)

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Advanced Professional Communication**		**06**
	Define advanced workplace communication concepts and models. Analyze workplace communication scenarios and conflict situations. Demonstrate emotional intelligence and ethical communication practices.		
	1.1	Fundamentals of Professional Communication: Types of professional communication, Barriers to communication, Principles of effective communication	
	1.2	Workplace Communication Skills: Formal and informal workplace communication, Internal and external communication, Corporate communication structure, Professional interaction techniques	
	1.3	Persuasive Communication and Influencing Skills: Persuasion techniques, Negotiation communication, Influencing strategies, Assertive communication	
	1.4	Emotional Intelligence, Interpersonal Skills and Professional Etiquettes: Ethical communication practices, Professional behavior and etiquette, Corporate responsibility in communication, Confidentiality and professionalism	
	Self-Learning Topics: *Watch corporate communication videos and prepare observation report*		
2	**Module 2: Technical and Business Writing**		**08**
	Draft technical reports, proposals, and executive summaries professionally. Prepare		

	professional emails, meeting agendas, and minutes. Apply concise and structured business writing principles.		
	2.1	Technical Report Writing: Structure of technical reports, Formal writing style, Data presentation techniques, Report formatting standards	
	2.2	Business Correspondence: Professional email writing, Business letters and memos, Escalation and response mails, Corporate communication drafting	
	2.3	Technical Proposal Writing: Proposal writing structure, Writing executive summaries, Parts of a technical proposal, Documentation standards	
	2.4	Business Meeting Documentation: Notice and agenda preparation, Writing meeting minutes, Documentation practices, Official communication records, Conducting business meetings	
	2.5	AI-Assisted Writing Tools: AI tools for professional writing, Content refinement techniques, Grammar and style checking tools, Ethical use of AI in documentation	
	Self-Learning Topics: *Create documentation using AI tools*		
3	**Module 3: Corporate Speaking & Group Communication**		**06**
	Deliver professional and impactful presentations confidently. Participate effectively in group discussions, panel discussions, and meetings. Demonstrate leadership and teamwork communication skills.		
	3.1	Professional Presentation Skills: Presentation structure, Visual aids and slide design, Voice modulation and body language, Audience engagement techniques	
	3.2	Group Discussions: GD strategies and participation, Analytical and critical discussion skills, Leadership during discussions, Professional interaction techniques	
	3.3	Team Communication and Collaboration: Team interaction methods, Collaborative communication, Workplace coordination, Communication in projects	
	3.4	Storytelling and Persuasive Speaking: Storytelling techniques, Persuasive presentation methods, Corporate storytelling, Influencing audience perception	
	3.5	Interview Communication Mastery: HR interview communication, Technical interview interaction, Behavioral interview handling, Mock interview practice	
	Self-Learning Topics: *Record and analyze personal presentation videos*		
4	**Module 4: Digital Communication & AI Tools**		**06**
	Use professional digital communication platforms effectively. Build professional online profiles and digital presence. Apply AI tools for presentations, reports, and communication tasks.		
	4.1	Digital Communication Platforms: Professional communication tools, Email and virtual communication, Online collaboration platforms, Digital workplace communication	
	4.2	LinkedIn and Professional Networking: LinkedIn profile optimization, Professional networking strategies, Building online professional identity, Networking etiquette	
	4.3	AI Tools for Communication: AI tools for presentations, AI-assisted report writing, Communication automation tools, Productivity enhancement using	

		AI	
	4.4	Digital Etiquette and Cyber Professionalism: Online professional behavior, Cyber ethics and safety, Responsible digital communication, Professional conduct in virtual environments	
	4.5	Personal Branding and Online Reputation: Personal branding strategies, Online presence management, Digital reputation building, Professional portfolio creation	
	Self-Learning Topics: *Build personal online portfolio*		
5	**Module 5: Placement and Career Readiness**		**06**
	Prepare ATS-friendly resumes and professional portfolios. Demonstrate effective communication during HR and technical interviews. Apply the STAR method while answering interview questions.		
	5.1	Resume and Portfolio Development: Resume writing techniques, ATS-friendly resume preparation, Cover letters, Portfolio building methods, Career documentation standards	
	5.2	Placement Communication Skills: Internship communication, Placement interaction strategies, Corporate communication readiness, Professional self-introduction	
	5.3	HR and Technical Interviews: HR interview preparation, Technical interview handling, Stress interview techniques, Behavioral interview strategies, STAR method for structured responses	
	5.4	Corporate Professionalism: Workplace expectations, Professional ethics and conduct, Office communication etiquette, Time and task communication	
	5.5	Career Growth and Workplace Readiness: Career planning communication, Workplace adaptability, Professional relationship management, Industry communication practices	
	Self-Learning Topics: *Watch placement interview sessions and prepare reflection report*		
6	**Module 6: Leadership, Innovation & Global Workplace Skills**		**04**
	Demonstrate leadership and innovation communication skills. Present entrepreneurial and technical ideas effectively. Handle crisis communication situations professionally.		
	6.1	Leadership Communication: Leadership communication styles, Motivational communication, Decision-making communication, Team leadership interaction	
	6.2	Innovation and Entrepreneurial Communication: Startup pitching techniques, Entrepreneurial communication, Innovation presentations, Business idea communication	
	6.3	Crisis Communication and Media Handling: Crisis management communication, Media interaction basics, Reputation management, Emergency communication strategies	
	6.4	Future Workplace Skills and Industry 5.0 Communication: Future communication trends, Human-AI workplace collaboration, Smart workplace communication, Industry 5.0 professional competencies	
	Self-Learning Topics: *Study successful startup presentations*		
		TOTAL	**36**

Suggested List of Activities

No.	Title and Details
1	Mock GD 1 **Objective: To develop students' ability to participate actively and effectively in group discussions.** **Outcome: Students will be able to present opinions logically and contribute meaningful ideas during group discussions.**
2	Business Meeting Simulation **Objective: To train students in professional communication and collaboration during formal business meetings.** **Outcome: Students will be able to participate professionally in meetings and demonstrate meeting etiquette.**
3	Mock GD 2 **Objective: To enhance students' confidence and communication skills in group discussion settings.** **Outcome: Students will be able to articulate viewpoints confidently and engage constructively in discussions.**
4	Presentation on Interpersonal Skills **Objective: To develop students' presentation and interpersonal communication skills.** **Outcome: Students will be able to organize and deliver structured presentations with effective verbal and non-verbal communication.**
5	Mock Interviews **Objective: To prepare students for professional interviews through practice and self-assessment.** **Outcome: Students will be able to answer interview questions confidently and communicate their strengths effectively.**
6	Creation of LinkedIn Profile **Objective: To guide students in building a professional online presence for career opportunities.** **Outcome: Students will be able to create a professional LinkedIn profile showcasing qualifications, achievements, and career interests.**
7	Debate on Emerging Technologies **Objective: To encourage critical thinking and effective public speaking through debate activities.** **Outcome: Students will be able to express arguments logically and defend viewpoints confidently.**
8	Pitching a Startup Idea **Objective: To enable students to present innovative business ideas persuasively and professionally.** **Outcome: Students will be able to explain a startup idea clearly, justify its feasibility, and deliver an effective pitch presentation.**
9	Final Group Discussion **Objective: To assess overall group discussion and communication skills.** **Outcome: Students will be able to participate confidently and demonstrate collaborative communication skills.**
10	Final Presentation on Project **Objective: To develop students' ability to present project work in a structured and**

	professional manner. **Outcome: Students will be able to deliver organized project presentations with clarity and effective audience engagement.**

Text Books

1. Business Communication Today – Bovee & Thill
2. Technical Communication – Meenakshi Raman

Reference Books

1. HBR Guide to Persuasive Presentations
2. Crucial Conversations – Patterson et al.
3. The Quick and Easy Way to Effective Speaking – Dale Carnegie

Assessment Methodology

Type of Assessment	Assessment Tools / Marks Distribution
Term Work (50 Marks)	Active Participation (Lab) = 05 marks | Assignments = 10 marks | Activities = 10 marks. Based on performance and satisfactory completion of all activities.
Internal Oral (25 Marks)	Book Report = 05 marks | Presentation = 10 marks | Final Group Discussion = 10 marks

7. Research Methodology

Course Code: 25CE5ELC | Category: ELC (Elective Learning Course)

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE5ELC	**Research Methodology**	2	-	-	2	-	-	**2**

Examination Scheme								
		CA	MSE	ESE	TW	OR	OR & PR	Total
Theory		-	-	-	50	-	-	**50**

Pre-Requisite Course(s)	25FE1BSE01 – Engineering Mathematics 01

Course Objectives & Module overview

Course Objectives:

This course introduces engineering students to the principles and practices of scientific research. It covers problem identification, literature review, research design, experimental methods, data analysis, technical writing, research ethics, and intellectual property rights. The course equips students with the skills required to undertake research projects, publish technical papers, and pursue innovation in engineering.

The objectives of this course are as follows:

1. Understand the fundamentals and methodologies of engineering research.
2. Formulate research problems and develop appropriate research designs.
3. Apply experimental, computational, and analytical methods in engineering investigations.
4. Analyze and interpret research data using statistical techniques.
5. Develop technical writing and presentation skills for dissemination of research findings.
6. Understand research ethics, intellectual property rights, and publication practices.

Module 1 introduces the fundamentals of engineering research, including its purpose, significance, and role in technological innovation. It covers the research process, scientific inquiry, and various types of research such as basic, applied, experimental, and computational research. Students will gain an understanding of how research contributes to solving engineering problems and advancing technology.

Module 2 focuses on identifying research problems and conducting effective literature reviews. It introduces techniques for searching, evaluating, and synthesizing scholarly literature to identify research gaps and formulate research objectives, questions, and hypotheses. Students will develop the ability to define meaningful engineering research problems based on existing knowledge and emerging challenges.

Module 3 introduces the principles of research design and methodology used in engineering research. It covers the selection of appropriate research approaches, methods, and procedures for conducting systematic investigations. Students will learn to design research studies, develop experimental plans, select data collection methods, and ensure the reliability and validity of research outcomes.

Module 4 focuses on methods of data collection, organization, and analysis in engineering research. It introduces primary and secondary data sources, sampling techniques, and statistical tools used for data interpretation. Students will learn to apply descriptive and inferential statistics to analyze research data, draw meaningful conclusions, and support evidence-based decision-making.

Module 5 introduces the ethical, legal, and professional aspects of engineering research. It covers research integrity, plagiarism, intellectual property rights (IPR), patents, copyrights, and scholarly publishing practices. Students will gain an understanding of ethical research conduct, protection of research outcomes, and the process of publishing technical papers in journals and conferences.

Module 6 focuses on the effective communication of research findings through technical writing and professional presentations. It covers the preparation of research papers, technical reports, theses, and research proposals, along with citation and referencing practices.

Course Outcomes

CO	Course Outcome
CO1	Explain the fundamental concepts, types, and processes of research in engineering and technology.
CO2	Identify research problems, formulate objectives and hypotheses, and conduct a comprehensive literature review using scientific databases.
CO3	Design appropriate research methodologies, experiments, and data collection procedures for engineering investigations.
CO4	Apply statistical and analytical techniques to collect, process, analyze, and interpret research data.
CO5	Demonstrate awareness of research ethics, plagiarism, intellectual property rights (IPR), and publication practices.
CO6	Prepare and present research proposals, technical reports, dissertations, and research papers in a professional format.

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Introduction to Engineering Research Concepts and Types of Research**		**04**
	After completing this module, students will understand the fundamentals and scope of engineering research.		
	1.1	Meaning, objectives, and significance of research; Types of research: Basic, Applied, Experimental, Computational, Action research; Characteristics of good research, Research process and scientific method, Engineering research challenges and opportunities	
	Self-Learning Topics: *Emerging Research Paradigms, Research Impact Assessment, Global Research Trends, AI in Engineering Research, Technology Transfer*		
2	**Module 2: Literature Review and Research Problem Formulation**		**04**
	After completing this module, students will be able to formulate research problems and conduct literature reviews.		

	2.1	Identification of research problems, Selection of research topics, Research objectives and questions, Hypothesis formulation, Research gap identification, Sources of technical literature, Literature survey techniques, Citation and referencing methods, Use of digital databases and search engines	
	Self-Learning Topics: *Research Question Development Frameworks, AI-Assisted Literature Reviews, Critical Analysis of Research Papers*		
3	**Module 3: Research Design and Methodology**		**08**
	After completing this module, students will be able to design suitable methodologies for engineering research.		
	3.1	Research design concepts, Quantitative and qualitative research methods, Experimental research design, Survey methods, Case study approach, Modeling and simulation-based research, Variables and measurement scales, Reliability and validity	
	Self-Learning Topics: *Bibliometric Analysis, AI-Assisted Research*		
4	**Module 4: Data Collection and Statistical Analysis**		**06**
	After completing this module, students will be able to collect and analyze engineering data.		
	4.1	Sources of data, Primary and secondary data, Sampling methods, Questionnaire design, Data coding and classification, Descriptive statistics, Probability distributions, Hypothesis testing, Correlation and regression analysis, ANOVA	
	Self-Learning Topics: *Data Ethics and Privacy, Open Data Repositories, Uncertainty Analysis, Perform exploratory analysis on an open dataset*		
5	**Module 5: Research Ethics, IPR, and Technical Publications**		**04**
	After completing this module, students will understand ethical and legal aspects of research.		
	5.1	Research ethics, Plagiarism and academic integrity, Ethical issues in engineering research, Intellectual Property Rights (IPR), Patents, copyrights, trademarks, Patent search and filing basics, Journal publications and conferences, Peer-review process	
	Self-Learning Topics: *Responsible Conduct of Research, Ethics in Emerging Technologies, Open Science Practices, Predatory Publishing, Research Visibility and Scholarly Identity*		
6	**Module 6: Technical Writing and Research Presentation**		**04**
	After completing this module, students will be able to communicate research findings effectively.		
	6.1	Scientific Writing, Research proposal writing and Funding Agencies, Thesis and dissertation preparation, Writing journal papers, Preparation of tables, figures, and references (IEEE, APA, MLA, Chicago, Harvard, Vancouver styles), Use of Reference Management Tools (Zotero, Mendeley, EndNote), Responding to Reviewers Comments, Research Communication through Blogs and social media	
	Self-Learning Topics: *Graphical Abstracts and Infographics, Digital Tools for Academic Writing, Peer Review and Manuscript Revision*		
		TOTAL	**30**

Text Books

1. Creswell, J.W. & Creswell, J.D. – Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, 6th Edition, SAGE Publications, 2023
2. Kothari, C. R. & Garg, G. – Research Methodology: Methods and Techniques, 5th Edition, New Age International Publishers, 2024
3. Ganesan, R. – Research Methodology for Engineers, MJP Publishers, 2024

Useful Links

1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_ge66
2. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ge79
3. https://ocw.mit.edu/courses/esd-932-engineering-ethics-spring-2006/
4. own Business

Assessment Methodology

Type of Assessment	Assessment Tools / Marks Distribution
Term Work (50 Marks)	Any two individual pedagogy assignments (10 marks each): Literature Review Assignment Statistical Analysis Using Software Patent Search Activity. Group activity/tasks (10 marks each): Technical Paper Review and Presentation Research Problem Identification Exercise Research Proposal Preparation

8. Mini Project III – Innovation, Research and Product Engineering

Course Code	Course Name	Teaching Scheme (Hrs./Week)			Credits Assigned			
		L	P	T	L	P	T	Total
25CE4CEP03	**Community Engagement Project III – Innovation, Research and Product Engineering**	-	1	-	-	1	-	**1**

Examination Scheme								
		CA	MSE	ESE	TW	OR	PR	Total
Theory		-	-	-	25	25	-	**50**

Pre-Requisite Courses	25CE4CEP01 – Community Engagement Project I (Design Thinking Fundamentals) 25CE4CEP02 – Community Engagement Project II (Solution Development and Testing)

Course Objectives & Overview

This course represents the culmination of a three-semester experiential learning journey, building upon 25CE4CEP01 – Mini Project I (Design Thinking Fundamentals) and 25CE4CEP02 – Mini Project II (Solution Development and Testing). Students are expected to transform validated ideas, prototypes, and challenge statements into impactful engineering solutions addressing industrial, societal, sustainability, research, entrepreneurship, and national innovation needs. Projects may originate from initiatives such as Smart India Hackathon (SIH), ISRO, government departments, industry collaborations, Sustainable Development Goals (SDGs), startup concepts, research problems, or community challenges.

The course emphasizes engineering innovation, solution maturity, experimentation, validation, deployment, and effective communication of outcomes. Students apply modern engineering practices, including software development, cloud deployment, DevOps, AI-assisted tools, testing methodologies, research techniques, and professional documentation, to develop, evaluate, and showcase deployable solutions with measurable impact.

Course Objectives (COs) – 5 Points

- To identify and analyze real-world problems from industrial, societal, research, or entrepreneurial domains and formulate clear project objectives, scope, and success criteria.
- To design innovative and feasible solutions by developing suitable architectures, selecting appropriate technologies, and planning validation and deployment strategies.
- To develop and implement project solutions using modern engineering practices, software tools, version control systems, APIs, cloud services, and AI-assisted technologies.
- To evaluate and validate project outcomes through systematic testing, performance analysis, user feedback, and assessment of scalability, sustainability, and impact.
- To effectively communicate and demonstrate project results through technical documentation, research reports, presentations, pitch decks, and professional project deployment or showcasing.

Course Outcomes

CO	Course Outcome
CO1	Identify and justify a significant industrial, societal, research, sustainability, or innovation challenge for solution development. (Remembering)
CO2	Interpret stakeholder requirements, literature findings, domain constraints, and problem-specific opportunities. (Understanding)
CO3	Apply appropriate engineering, research, and project management methodologies to develop innovative solutions. (Applying)
CO4	Analyze system performance, user feedback, experimental results, and feasibility constraints to improve solution quality. (Analyzing)
CO5	Evaluate the impact, scalability, sustainability, novelty, and effectiveness of the proposed solution. (Evaluating)
CO6	Develop and demonstrate a deployable product, research prototype, competition-ready solution, or innovation artifact with professional documentation and presentation. (Creating)

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Project Identification and Problem Framing**		**06**
	After completing this module, students will be able to: Identify real-world, industrial, societal, or research challenges; Formulate project objectives and scope; Establish measurable success criteria; Define project deliverables.		
		Problem sources: SIH, ISRO, SDGs, industry, startups, community. Problem analysis, feasibility assessment, stakeholder identification. Project planning, milestones, risk identification.	
	Deliverable	Project Proposal: Problem statement, objectives, literature review, project plan	
2	**Solution Design and Technical Planning**		**06**
	After completing this module, students will be able to: Design solution architecture; Select suitable technologies and methodologies; Plan validation and testing strategies; Estimate resources and deployment requirements.		
		Solution architecture and workflow design. Technology selection and validation strategy. Resource estimation and deployment planning.	
	Deliverable	Design Document: Architecture, workflow, technology stack, testing strategy	
3	**Development, Experimentation and Engineering Practices**		**08**
	After completing this module, students will be able to: Develop solution components; Apply software engineering practices; Integrate external services and tools; Manage project versions and releases.		
		Core solution development and integration. Version control, documentation, engineering practices. AI-assisted tools, cloud services, APIs.	

	Deliverable	Working Prototype: Source code repository, implementation progress, demo	
4	**Testing, Validation and Impact Assessment**		**06**
	After completing this module, students will be able to: Validate developed solutions; Measure performance and effectiveness; Assess user acceptance and impact; Improve solution quality through iteration.		
		Functional, usability, performance, and security testing. Experimental validation and benchmarking. Impact, scalability, and sustainability assessment.	
	Deliverable	Validation Report: Test results, user feedback, performance/impact analysis	
5	**Product Deployment, Research Communication and Pitching**		**04**
	After completing this module, students will be able to: Deploy or demonstrate solutions professionally; Prepare research and technical documentation; Communicate innovation outcomes effectively; Present solutions to academic, industry, or competition panels.		
		Deployment and demonstration. Research paper and technical report preparation. Pitch deck and presentation techniques.	
	Deliverable	Final Submission Package: Technical report, presentation, deployed solution/prototype, research paper/pitch deck/SIH submission	

Suggested Project Categories

- Smart India Hackathon (SIH) Problem Statements
- ISRO / Government Innovation Challenges
- Sustainable Development Goal (SDG) Projects
- Research and Innovation Projects
- Product Engineering and Startup Ideas
- AI and Data-Driven Applications
- Human-Machine Interaction Solutions
- Cloud-Based Applications
- Community and Social Impact Projects
- Industry-Sponsored Projects

Assessment Criteria

- Problem significance and relevance
- Technical depth and innovation
- Application of engineering/research methodologies
- Quality of implementation or experimentation
- Validation and impact assessment
- Deployment or demonstration readiness
- Research paper and documentation quality
- Presentation and pitching effectiveness
- Teamwork and project management
- Individual contribution and professional conduct

Text Books

1. Tim Brown – Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation, Harper Business, 2009

2. Harold Kerzner – Innovation Project Management: Methods, Case Studies, and Tools for Managing Innovation Projects, Wiley, 2nd Edition, 2023
3. Gerard O'Regan – Guide to Software Project Management, Springer, 2025
4. Jeanne Liedtka, Randy Salzman, Daisy Azer – Design Thinking for the Greater Good: Innovation in the Social Sector, Columbia University Press, 2017

Reference Books

1. Jim Highsmith – Agile Project Management: Creating Innovative Products, Addison-Wesley
2. Roger L. Martin – The Design of Business: Why Design Thinking is the Next Competitive Advantage, Harvard Business Review Press
3. Frederick P. Brooks Jr. – The Mythical Man-Month: Essays on Software Engineering, Addison-Wesley
4. David Kelley and Tom Kelley – Creative Confidence: Unleashing the Creative Potential Within Us All, Crown Business

Useful Links

1. Smart India Hackathon (SIH) Official Portal: https://www.sih.gov.in
2. ISRO Capacity Building and Student Programmes: https://www.isro.gov.in/CapacityBuilding.html
3. United Nations Sustainable Development Goals (SDGs): https://sdgs.un.org/goals
4. Stanford d.school Resources and Design Tools: https://dschool.stanford.edu/resources
5. IDEO Design Thinking Resources: https://designthinking.ideo.com/resources
6. GitHub Student Developer Resources: https://education.github.com/students
7. IEEE Author Center and Paper Templates: https://journals.ieeeauthorcenter.ieee.org

Assessment Methodology

Type of Assessment	Assessment Tools / Marks Distribution
TW (25 Marks)	Active Participation (Lab) = 05 marks | Project Report = 10 marks | Progress Presentations (Minimum 2) & Demonstration = 10 marks
PR (25 Marks)	Practical examination based on experiments performed by the students during laboratory sessions.

9. Mathematics for Data Science

Course Code: HDSC501 | Program: Data Science (TE Sem V)

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tutorial	Total
HDSC501	**Mathematics for Data Science**	04	--	--	04	--	--	**04**

Examination Scheme							
	CA	MSE	ESE	TW	OR	OR 7 PR	Total
HDSC501 – Mathematics for Data Science	20	30	50			--	**100**

Pre-Requisite Course(s)	Applied Mathematics | Discrete Mathematics	

Course Objectives & Module Overview

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Course Objectives

This course introduces students to the mathematical foundations required for data science and analytics. It covers linear algebra, probability and statistics, data visualization, exploratory data analysis, optimization techniques, and dimensionality reduction methods. The course equips students with the mathematical and analytical skills necessary for understanding, modeling, and solving data-driven problems in engineering and business applications.

The objectives of this course are as follows:

1. Understand the fundamental mathematical concepts that support data science and analytics.
2. Apply linear algebra, probability, and statistical techniques to model and analyze real-world data.
3. Develop skills in graphical representation and visualization of quantitative and qualitative data.
4. Perform exploratory data analysis to extract meaningful insights and support decision-making.
5. Analyze optimization methods used in data-driven problem solving.
6. Understand and apply dimensionality reduction techniques for efficient data analysis and machine learning applications.

Module 1: Linear Algebra

Module 1 introduces the fundamental concepts of linear algebra that form the basis of modern data science. It covers vectors, matrices, systems of linear equations, fundamental subspaces, eigenvalues,

eigenvectors, and singular value decomposition (SVD). Students will learn mathematical techniques used in data representation, transformation, and machine learning algorithms.

Module 2: Probability and Statistics

Module 2 focuses on probability theory and statistical concepts essential for data analysis. It introduces random variables, probability distributions, random sampling, sampling distributions, Chi-square, t-, and F-distributions, bivariate normal distributions, and the Central Limit Theorem. Students will develop the ability to model uncertainty and perform statistical inference in practical applications.

Module 3: Introduction to Graphs and Data Visualization

Module 3 introduces various types of quantitative and qualitative data and appropriate graphical methods for their representation. It covers bar charts, pie charts, histograms, stem-and-leaf plots, dot plots, scatter plots, time-series graphs, and other graphical techniques. Students will learn to select suitable visualization methods for effective communication and interpretation of data.

Module 4: Exploratory Data Analysis

Module 4 focuses on exploratory data analysis (EDA) techniques used for understanding and preparing datasets. It covers data cleaning, handling missing values, feature engineering, visualization, and interpretation of patterns and trends in data. Students will learn how EDA supports informed decision-making and the selection of suitable machine learning models.

Module 5: Optimization Techniques

Module 5 introduces optimization concepts used in data science and engineering applications. It covers constrained and unconstrained optimization problems along with numerical optimization methods such as the Bisection Method and False Position Method. Students will gain an understanding of optimization approaches for improving analytical models and decision-making processes.

Module 6: Dimensionality Reduction Algorithms

Module 6 focuses on dimensionality reduction techniques used to simplify high-dimensional datasets. It introduces linear methods such as Principal Component Analysis (PCA), Factor Analysis, and Linear Discriminant Analysis (LDA), as well as non-linear techniques including Multidimensional Scaling (MDS) and Isometric Feature Mapping (ISOMAP). Students will learn how dimensionality reduction improves visualization, computational efficiency, and predictive model performance.

Course Outcomes

CO	Course Outcome
CO1	**Recall** the fundamental concepts of linear algebra, probability, statistics, optimization, and dimensionality reduction used in data science.
CO2	**Explain** the principles of probability distributions, sampling distributions, graph representations, and exploratory data analysis techniques.
CO3	**Apply** linear algebra, probability, and statistical methods to solve data-driven problems and perform data analysis.
CO4	**Analyze** datasets using appropriate visualization techniques and exploratory data analysis methods to identify patterns and relationships.

CO5	**Evaluate** optimization methods and dimensionality reduction techniques for their suitability in data analytics applications.
CO6	**Develop** data-driven solutions by integrating visualization, optimization, and dimensionality reduction techniques for effective decision-making and model improvement.

Syllabus

Module No.	Unit No.	Topics	Hours
1	**Module 1: Linear Algebra**		**05**
	1.1	Vectors and Matrices, Solving Linear Equations, The Four Fundamental Subspaces, Eigenvalues and Eigenvectors, The Singular Value Decomposition (SVD).	
2	**Module 2: Probability and Statistics**		**09**
	2.1	Introduction, Random Variables and their Probability Distribution, Random Sampling, Sample Characteristics and their Distributions, Chi-Square, t-, and F-Distributions: Exact Sampling Distributions, Sampling from a Bivariate Normal Distribution, The Central Limit Theorem.	
3	**Module 3: Introduction to Graphs**		**10**
	3.1	Quantitative vs. Qualitative data, Types of Quantitative data: Continuous data, Discrete data, Types of Qualitative data: Categorical data, Binary data, Ordinary data, Plotting data using Bar graph, Pie chart, Histogram, Stem and Leaf plot, Dot plot, Scatter plot, Time-series graph, Exponential graph, Logarithmic graph, Trigonometric graph, Frequency distribution graph.	
4	**Module 4: Exploratory Data Analysis**		**09**
	4.1	Need of exploratory data analysis, Cleaning and preparing data, Feature engineering, Missing values, Understand dataset through various plots and graphs, Draw conclusions, Deciding appropriate machine learning models.	
5	**Module 5: Optimization Techniques**		**10**
	5.1	Types of optimization – Constrained and Unconstrained optimization, Methods of Optimization – Numerical Optimization, Bracketing Methods: Bisection Method, False Position Method.	
6	**Module 6: Dimension Reduction Algorithms**		**05**
	6.1	Introduction to Dimension Reduction Algorithms, Linear Dimensionality Reduction: Principal Component Analysis (PCA), Factor Analysis, Linear Discriminant Analysis (LDA).	
	6.2	Non-Linear Dimensionality Reduction: Multidimensional Scaling, Isometric Feature Mapping (Isomap), Minimal Polynomial.	

TOTAL	**48**

Text Books

1. Gilbert Strang – Linear Algebra for Everyone, Wellesley Cambridge Press
2. Vijay Rohatgi – An Introduction to Probability and Statistics, Wiley Publication
3. Edwin Chong, Stanislaw Zak – An Introduction to Optimization, 2nd Edition, Wiley
4. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong – Mathematics for Machine Learning, Cambridge University Press
5. John Tukey – Exploratory Data Analysis, Princeton University and Bell Laboratories

Reference Books

1. Gilbert Strang – Introduction to Linear Algebra
2. Erwin Kreyszig – Advanced Engineering Mathematics
3. Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar – Foundations of Machine Learning, MIT Press, 2018
4. Shai Shalev-Shwartz and Shai Ben-David – Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press, 2014
5. William B. Claster – Mathematics and Programming for Machine Learning with R, CRC Press, 2020

Useful Links

1. https://math.mit.edu/~gs/linearalgebra/
2. https://www.coursera.org/learn/probability-theory-statistics
3. https://nptel.ac.in/courses/111/105/111105090/
4. https://onlinecourses.nptel.ac.in/noc21_ma01/preview
5. https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures/